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A Hybrid Intelligent Optimization Algorithm Combining Weighted K-means Clustering and Improved ABC for Optimal Deployment of Prehospital Emergency Heliports

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ABSTRACT

Prehospital emergency helicopters, with excellent maneuverability and response speed, have become the key equipment for improving the quality of emergency medical services. However, the scientific deployment of heliports directly determines the efficiency and cost of the rescue network. To solve this complex optimization problem, this paper proposes a hybrid intelligent optimization algorithm that combines weighted K-means clustering and improved artificial bee colony (WKM-IABC). Firstly, depending on the severity of patient injury, differentiated weights are assigned to the demand points, and the weighted K-means algorithm is used to determine high-quality initial heliport locations. Secondly, considering the three primary objectives of service coverage ratio, rescue response time and economic cost, the Analytic Hierarchy Process (AHP) is applied to quantify weighted preferences and construct a multi-objective integrated optimization model. Finally, an improved ABC (IABC) algorithm is designed to optimize the heliport locations accurately and efficiently through global optimal guidance, dynamic selection strategy, and high-quality initialization. The simulation experiment results demonstrate that compared with state-of-the-art optimization algorithms, the proposed WKM-IABC algorithm exhibits significant advantages in comprehensive fitness, service coverage ratio, convergence speed, and robustness. Statistical significance tests further verify the superiority of WKM-IABC. This study will provide a more reasonable and feasible deployment scheme for prehospital emergency heliports.

Keywords: Weighted K-means, Artificial bee colony, Prehospital emergency helicopters, Deployment scheme, Hybrid intelligent optimization

1. Introduction

Prehospital emergency, as a critical element of the emergency medical service system, significantly influences the therapeutic efficacy and survival rates of critically ill patients through response efficiency[1–3]. For patients with severe trauma or acute myocardial infarction, it is crucial to expedite the rescue response efficiency[4,5]. However, due to ground transport congestion and complex geographical constraints, conventional ambulances often find it difficult to arrive at the scene within the optimal time window[6]. Therefore, prehospital emergency helicopters, leveraging their unique aerial maneuverability, can effectively avoid ground obstacles and achieve rapid point-to-point access[7,8]. This capability positions them as one of the important symbols of advanced medical rescue systems in developed countries. The outstanding performance of prehospital emergency helicopters in remote areas, major traffic incidents, and natural disasters fully underscores their significance in enhancing regional emergency support capabilities[9–11].

Despite the advantages of prehospital emergency helicopters, their operational costs and maintenance costs are the primary constraints on their broad application[12]. Compared to developed countries, the helicopter rescue system in developing countries still encounters numerous challenges, including infrastructure deficiencies and a lack of scientific planning[13]. The construction of an efficient rescue network necessitates not only enhance the performance of helicopters themselves but also the reasonable deployment of ground infrastructure such as heliports[14]. Inefficient heliport site selection can lead to the blind spots of service coverage, delayed rescue response efficiency, and significant resource idle and waste. Therefore, the scientific planning of heliport locations and quantities, within budgetary limitations, to optimize rescue response efficiency and maximize service coverage ratio, is a critical and urgent research topic for currently promoting the development of the aviation medical rescue industry.

In the field of prehospital emergency services, the scientific deployment of heliports constitutes a critical facility location problem[15]. Numerous studies have conducted many explorations, including coverage models[16], P-median models[17] and P-center models[18]. While these conventional methods provide valuable theoretical frameworks, they often lack sufficient consideration for the complex geographical constraints, dynamic demand distribution and trade-offs between multi-objectives in the practical problem. In recent years, genetic algorithm (GA), particle swarm optimization algorithm (PSO) and others have gained prominence in tackling these complex spatial optimization challenges, leveraging their robust global search capabilities[19–21]. However, these algorithms still have significant room for enhancement in solution quality and convergence speed.

In addition, clustering-based hybrid optimization and feature selection strategies have become research hotspots in the field of intelligent optimization in recent years[22]. An augmented weighted K-means grey wolf optimizer was proposed for data clustering problems, which verifies the effectiveness of combining clustering with intelligent optimization[23]. A novel clustering-based hybrid feature selection approach using ant colony optimization further improves the feature extraction capability of clustering algorithms by introducing ant colony optimization[24]. Enhancing K-means Clustering Performance with a Two-Stage Hybrid Preprocessing Strategy optimizes the preprocessing link of K-means and improves the clustering accuracy[25]. However, few studies apply the latest clustering-based hybrid optimization strategies to the deployment of prehospital emergency heliports, and there is a lack of combination of demand point feature weighting and intelligent optimization algorithm improvement. Therefore, there is a compelling need to develop an algorithm that can more effectively balance the complex relationship between service coverage ratio, rescue response time and economic cost.

According to the current research, this paper proposes a hybrid intelligent optimization algorithm that combines weighted K-means clustering and improved artificial bee colony (WKM-IABC) for optimal deployment of prehospital emergency heliports. The primary contributions of this work are as follows:

(1) Data-driven weighted initialization innovation: For the first time, the severity of prehospital emergency patients is quantified as clustering weights, breaking the limitation of equal weight clustering in traditional K-means. The initial heliport location is more in line with the actual emergency demand, solving the problem of low initial solution quality caused by traditional random initialization or non-weighted clustering. This weighting strategy is combined with the actual clinical emergency grading standards, making the initial solution have practical engineering significance rather than just theoretical optimization.

(2) Multi-dimensional improvement innovation of ABC algorithm: The combination of global optimal guided honey source update, sorting-based dynamic selection, and weighted K-means-driven high-quality population initialization three improvement strategies breaks the defects of slow convergence and easy to fall into local optimum of traditional ABC algorithm. The three strategies work together to realize the optimization effect of accurate initial, directional search and elite mining, and this improvement combination is applied for the first time in the heliport deployment problem, which effectively improves the algorithm's optimization performance for spatial facility location problems.

(3) Emergency scenario-oriented multi-objective optimization model innovation: The weight determination method based on AHP fully integrates the actual decision preferences of emergency management departments, and scientifically aggregates the three conflicting objectives of coverage rate, response time and economic cost. It breaks the problem of high decision difficulty of traditional Pareto optimization in practical application, and the model design is more in line with the engineering practice of urban prehospital emergency, and the optimization results have direct guiding significance for the actual deployment of heliports.

This study aims to provide a scientific, feasible, and cost-effective decision support tool for the deployment of prehospital emergency heliports.

2. Materials and Methods

2.1 Weighted K-means Algorithm

The K-means algorithm is a classic unsupervised clustering method for data partitioning, whose core goal is to partition a dataset $D = \{X_1, X_2, \cdots, X_n\}$into K disjoint clusters, such that the similarity of sample data within each cluster is the highest and the similarity between clusters is the lowest[26].

Firstly, K initial cluster centers are specified to divide the given dataset into K clusters. Then, the algorithm continuously minimizes the Sum of Squared Error (SSE) of each cluster through multiple iterative updates of cluster centers. Clustering ends when the cluster centers and the cluster to which each data point belongs tend to be stable, and the final clustering result is obtained[27]. The specific formula is as follows:

$$SSE = \sum_{i=1}^{K} \sum_{X \in S_i} \|X - \mu_i\|^2 \tag{1}$$

where K is the number of cluster centers. $S_i \in S, S = \{S_1, S_2, \cdots, S_K\}$ is the set composed of all clusters. $\|X - \mu_i\|^2$ represents the square of the Euclidean distance between X and $\mu_i \in \mu, \quad \mu = \{\mu_1, \mu_2, \ldots, \mu_K\}$ are all the center points. The smaller the SSE value, the better the clustering effect, indicating that the data points within the cluster are more concentrated around the cluster center.

In the traditional K-means algorithm, each data point contributes equally to the clustering result, which is not in line with the actual demand of prehospital emergency heliport deployment—different patient demand points have different importance due to the difference in injury severity. To address this issue, the weighted K-means algorithm is adopted in this paper, which has received widespread attention and research in practical clustering problems.

In the weighted K-means algorithm, each data point is assigned a weight value based on practical problem characteristics, which can accurately reflect the importance of the data point in the clustering process. Similarly, the weighted K-means algorithm determines the final cluster centers by minimizing the weighted Sum of Squared Error (WSSE). The specific formula is as follows:

$$WSSE = \sum_{i=1}^{K} \sum_{X \in S_i} \omega_i \|X - \mu_i\|^2 \tag{2}$$

where ω_i is the weight value of the i-th data point, which is determined by the severity of patient injury in this paper. $\omega_i \|X - \mu_i\|^2$ represents the weighted value of the squared Euclidean distance between data point X and μ_i.

The weighted K-means algorithm iteratively assigns each data point to the nearest weighted cluster center and updates the weighted cluster center until convergence. In this process, the differences in the importance of different data points are fully considered, which improves the accuracy and credibility of the clustering results, thus having broad application prospects in practical optimization problems such as emergency facility location.

2.2 Elbow Method and Silhouette Coefficient

In clustering algorithms represented by K-means, the number of cluster centers K needs to be given in advance, which is a key parameter that directly affects the clustering effect. The common methods for determining the optimal K value include the empirical method, elbow method and silhouette coefficient[28]. The empirical method relies on historical experience or basic domain knowledge to determine the number of cluster centers, which can usually be adopted in situations where accuracy requirements are not high. However, when using the weighted K-means algorithm, the assignment of different weights to each data point based on its importance leads to an increase in data dimensionality and clustering complexity, thereby necessitating a higher level of accuracy in the determination of the K value[29]. This paper adopts a combined method of elbow method and silhouette coefficient to determine the optimal K value, taking advantage of their complementary strengths to improve the accuracy of K value selection.

2.2.1 Elbow Method

As the number of cluster centers K increases, the SSE value of the clustering result typically shows a decreasing trend. This is because more cluster centers mean that data points in each cluster are closer to their corresponding cluster centers. The elbow method is to calculate the SSE values under different K values and plot the SSE-K curve[30]. The optimal clustering center number K can be determined by finding the "elbow point" where the SSE value begins to decrease at a significantly slower rate. Essentially, before this point, increasing the number of cluster centers can substantially reduce the SSE value and significantly improve the clustering effect; after this point, the increase in the number of cluster centers no longer has a significant impact on the SSE value. Therefore, the K value corresponding to the elbow point is the optimal number of cluster centers.

2.2.2 Silhouette Coefficient

The silhouette coefficient is a commonly used clustering performance evaluation index that combines the intra-cluster density and inter-cluster separation degree of clustering results to provide a quantitative evaluation value for each sample point, with a value range from -1 to 1[31]. The closer the silhouette coefficient value is to 1, the better the clustering effect, indicating that the sample point is highly similar to the internal points of its cluster and highly dissimilar to the points of other clusters; the closer the value is to -1, the worse the clustering effect, indicating that the sample point is more suitable to be divided into other clusters; a value close to 0 indicates that the sample point is on the boundary of two clusters.

Firstly, calculate the silhouette coefficient of all data points under different K values, including the average intra-cluster distance a_i (the average dissimilarity between the i-th data point and other data points in the same cluster) and the average inter-cluster distance b_i (the minimum average dissimilarity between the i-th data point and all data points in other clusters). Then, calculate the silhouette coefficient of each data point and the average silhouette coefficient of the entire dataset. The specific calculation formulas are as follows:

$$S_i = \frac{b_i - a_i}{\max\{a_i, b_i\}} \tag{3}$$

$$S = \text{average}\{S_1, S_2, \ldots, S_n\} \tag{4}$$

where S_i represents the silhouette coefficient of the i-th data. S represents the average silhouette coefficient of the entire dataset. By drawing the average silhouette coefficient-K curve, the K value with the highest average silhouette coefficient is selected as the optimal number of clustering centers.

Both the elbow method and the silhouette coefficient have certain limitations in practical application: the elbow method requires subjective judgment in identifying the "elbow point", and the judgment result may have certain differences for different researchers; the application of the silhouette coefficient needs to consider the balance and interpretability of clustering, and it is not advisable to select the K value only based on the highest silhouette coefficient. Therefore, this paper integrates both methods for optimal K value selection, which can effectively make up for the deficiencies of a single method and improve the scientificity of the optimal number of heliports determination.

2.3 IABC Algorithm

2.3.1 Traditional ABC Algorithm

ABC algorithm is an intelligent optimization algorithm proposed by Karaboga et al. based on the efficient foraging behavior of honey bee colonies[32]. The core idea of the algorithm is to map the feasible solutions of the optimization problem to the honey sources in the search space, and the quality of the feasible solutions is corresponding to the nectar amount of the honey sources. During the search process, different types of artificial bees work together to find high-quality honey sources that meet the requirements, and continuously optimize the quality of the solutions through information exchange and sharing between bees.

In the ABC algorithm, artificial bees are divided into three types according to their different functions and division of labor: employed bees, onlooker bees, and scout bees[33]. Employed bees are responsible for exploring the neighborhood of the honey sources they occupy, generating new honey sources, and sharing the nectar amount information of the honey sources with onlooker bees with a certain probability; onlooker bees select high-quality honey sources for further neighborhood search based on the information shared by employed bees, and optimize the honey source locations as much as possible; scout bees are responsible for random search in the entire search space to discover new potential honey sources, which can effectively avoid the algorithm falling into local optima.

The implementation process of the traditional ABC algorithm mainly includes five stages: initialization stage, employed bee stage, onlooker bee stage, scout bee stage, and iterative condition assessment stage[34]. Initialization stage: similar to other optimization algorithms, the ABC algorithm first initializes the basic parameters of the algorithm, including the number of honey clusters, local maximum search times and iteration termination conditions[35]. Employed bee stage: the employed bee population is determined and partitioned according to the initialization stage and iteration conditions. Each employed bee launches a global search for a new feasible solution, and determines whether to replace the initial solution relying on the fitness. Onlooker bee stage: this stage mainly guides the search direction according to the honey source information provided by the employed bee population to mitigate irrelevant optimization processes. Scout bee stage: if the stagnation times of employed bees or onlooker bees reach the upper limit, re-initialize randomly, and reset the stagnation times, otherwise the algorithm will automatically bypass this phase. Iterative condition assessment stage: the optimal solution of the iteration process is updated and recorded according to the greedy criterion. If the iteration termination condition is satisfied at this point, the algorithm will terminate and output the result.

2.3.2 Improvement Strategy

In response to the potential shortcomings of poor initial solution quality, slow convergence speed and susceptibility to local optima of the traditional ABC algorithm when solving complex spatial optimization problems, this paper proposes three core improvement strategies aimed at enhancing the performance in the prehospital emergency heliports.

- **Global Optimal Guided Honey Source Update Mechanism**

  In the employed bee stage of traditional ABC algorithm, new honey sources are generated by random perturbations near the current honey source, and their exploration direction has a certain degree of blindness. To accelerate algorithm convergence and utilize the optimal information discovered by the population, this paper introduces a directed learning mechanism. Specifically, when generating candidate honey sources, a probability threshold is set. This study conducted preliminary parameter sensitivity analysis and set it to 0.7. If the random number is less than the given probability threshold, the new honey source will be searched around the current global optimal honey source, and the update process is as follows:

$$\mu_{\text{new}} = \mu_{\text{best}} + \alpha \times (\mu_{\text{best}} - \mu_i) \tag{5}$$

where μ_{new} represents the new honey source solution generated by the update, μ_{best} is the current global optimal honey source solution, μ_i is the original honey source solution, α is a random number within the interval of $[-1,1]$. This strategy allows the population to have a higher probability of conducting fine searches in known optimal regions, effectively guiding the search direction and accelerating convergence speed.

On the contrary, if the random number is greater than the given probability threshold, the random neighborhood search strategy of traditional ABC algorithm is used to maintain population diversity and balance the exploration capabilities.

- **Sorting-Based Dynamic Selection Strategy**

During the onlooker bee stage, the traditional ABC adopts roulette wheel selection, and its selection probability is directly calculated by normalizing the fitness of the honey source. However, in the continuous iteration process, the population fitness tends to be similar, resulting in the insufficient selection pressure. Therefore, this paper introduces a sorting-based exponential scaling method to recalculate the selection probability:

$$P_i = \frac{e^{-\alpha \cdot \text{rank}_i / SN}}{\sum_{j=1}^{SN} e^{-\alpha \cdot \text{rank}_j / SN}} \tag{6}$$

where $rank_i$ is the sequential number of all current honey sources sorted in descending order based on their fitness (optimal solution $rank = 1$), SN is the number of honey sources, α is the selection pressure coefficient (set as 4.0 in this study).

TThis formula ensures an exponentially increasing selection probability for high fitness honey sources, significantly enhancing the selection pressure of the algorithm, and allowing onlooker bees to focus more on elite solution areas for deep mining, thereby increasing the likelihood of finding high-quality solutions.

- **High-Quality Population Initialization Method Relying on Weighted K-means**

The traditional ABC algorithm usually initializes the honey source population randomly in the search space, but the quality and distribution of the randomly generated initial solutions cannot be guaranteed. Many low-quality initial solutions will lead to long training time and slow convergence speed of the algorithm, and even make the algorithm fall into local optima and fail to find the global optimal solution. To address this issue, this paper combines the weighted K-means clustering algorithm proposed in Section 2.1 and designs a high-quality population initialization method relying on weighted K-means. The core idea is to use the clustering results of the weighted K-means algorithm to generate the initial honey source population, so as to improve the quality and rationality of the initial solution. This initialization method makes the initial honey source population have a high quality and reasonable distribution, which can effectively reduce the number of iterations required for the algorithm to converge, improve the convergence speed of the algorithm, and lay a good foundation for the subsequent global optimization search of the algorithm.

In summary, the three improvement strategies work together, providing a superior search starting point for the algorithm, guiding the search to the most promising area during the iteration process, ensuring that search resources are concentrated on elite solutions. The combined effect of these three factors enables IABC algorithm to achieve faster convergence speed, stronger local development capability, and higher global optimization success rate when solving the heliport locations problem in this study.

2.4 Decision Preferences-Based Multi-Objective Integrated Optimization Model

The deployment of prehospital emergency heliports is essentially a typical multi-objective optimization problem, with core objectives including maximizing service coverage ratio, minimizing rescue response time, and controlling economic cost. These goals often conflict with each other. For example, increasing the number of heliports can improve coverage and potentially shorten response time, but increase the economic cost. To solve this multi-objective optimization problem, there are mainly two paradigms in previous researches: one is to generate a set of Pareto optimal solutions for posterior selection; the other one is to aggregate the multi-objectives into a single-objective function for optimization depending on the decision preferences[36]. Considering the application background of this study, we need to clarify the preferences of urban emergency management departments for each goal in the early planning. Therefore, we transform the multi-objective problem into a comprehensive single-objective optimization problem relying on the linear weighted sum method[37].

2.4.1 Objective Function

Firstly, this paper independently defines three sub-objective functions for the three core optimization objectives of heliport deployment, and unifies all sub-objective functions into maximization problems with a value range of [0,1] through normalization processing, which is convenient for the subsequent construction of the comprehensive fitness function.

- **Service Coverage Ratio Sub-objective Function**

  The service coverage ratio is the core optimization objective of prehospital emergency heliport deployment, which measures the proportion of patient demand points that can be covered by helicopters within the specified service radius. Considering the different weights of demand points due to the difference in patient injury severity, the weighted service coverage ratio is adopted as the sub-objective function, and the specific formula is as follows:

$$f_{cov} = \frac{\sum_{j=1}^{M} [\![\min(\min_{i=1}^{N} d_{ij}, R) \leq R]\!] \cdot \omega_j}{\sum_{j=1}^{M} \omega_j} \tag{7}$$

  where M is the number of data, N is the number of heliports, ω_j is the weight of the j-th data. d_{ij} is the Euclidean distance from the i-th data to the j-th heliport. $[\![\cdot]\!]$ is an indicate function. R is the helicopter service radius. When the condition is true, the value is 1, otherwise it is 0.

  The value range of f_{cov} is [0,1], and the larger the value, the higher the weighted service coverage ratio, indicating that more high-weight patient demand points are covered by the helicopter service range.

- **Rescue Response Time Sub-objective Function**

  The rescue response time measures the average flight time of helicopters from the nearest heliport to all patient demand points, which is a key indicator to ensure the rescue efficiency of prehospital emergency helicopters. Considering the different weights of demand points, the weighted average rescue response time is adopted, and the reciprocal form is used to transform it into a maximization problem. The specific formula is as follows:

$$f_{time} = \frac{1}{1 + \sum_{j=1}^{M} \frac{\min_{i=1}^{N} d_{ij}}{v} \cdot \omega_j} \tag{8}$$

  where v represents the average flight speed of helicopter, $\min_{i=1}^{N} d_{ij}$ represents the distance from j-th data to the nearest heliport. This function measures the weighted average flight time from the nearest heliport to all demand points.

  The value range of f_{time} is [0,1], and the larger the value, the shorter the weighted average rescue response time, indicating that the helicopter can arrive at the accident scene more quickly to carry out rescue work.

- **Economic Cost Sub-objective Function**

  The economic cost of heliport deployment includes the fixed construction cost of heliports and the variable operating cost of helicopter rescue (such as flight cost, maintenance cost and manpower cost). To control the economic cost on the premise of meeting the emergency demand, the reciprocal form is used to transform the economic cost into a maximization problem, and the specific formula is as follows:

$$f_{cost} = \frac{1}{1 + C_{fix} \times N + C_{variable}} \tag{9}$$

  where C_{fix} is the fixed construction cost of single heliport, C_{var} is the other variable cost. The total cost includes fixed construction costs and variable operating costs.

  The total economic cost of heliport deployment is defined as $W = C_{fix} \times N + C_{var}$. The value range of f_{cost} is [0,1], and the larger the value, the lower the total economic cost of heliport deployment, indicating that the resource utilization efficiency is higher.

- **Comprehensive Single-fitness Function**

  Based on the three normalized sub-objective functions, the linear weighted sum method is used to construct the comprehensive single-fitness function of WKM-IABC, which is used to evaluate the quality of the heliport deployment scheme. The specific formula is as follows:

$$\max F = \omega_1 \cdot f_{cov} + \omega_2 \cdot f_{time} + \omega_3 \cdot f_{cost} \tag{10}$$

where $0 \leq f_{\text{coverage}}, f_{\text{time}}, f_{\text{cost}} \leq 1, \quad \omega_1 + \omega_2 + \omega_3 = 1$. F is the comprehensive fitness value, with a value range of [0,1], and the larger the value, the better the heliport deployment scheme.

The comprehensive fitness function F is the core evaluation index of WKM-IABC, which integrates the three conflicting objectives of service coverage ratio, rescue response time and economic cost according to decision preferences, and is used as the fitness function for the honey source solution evaluation and greedy selection in IABC algorithm.

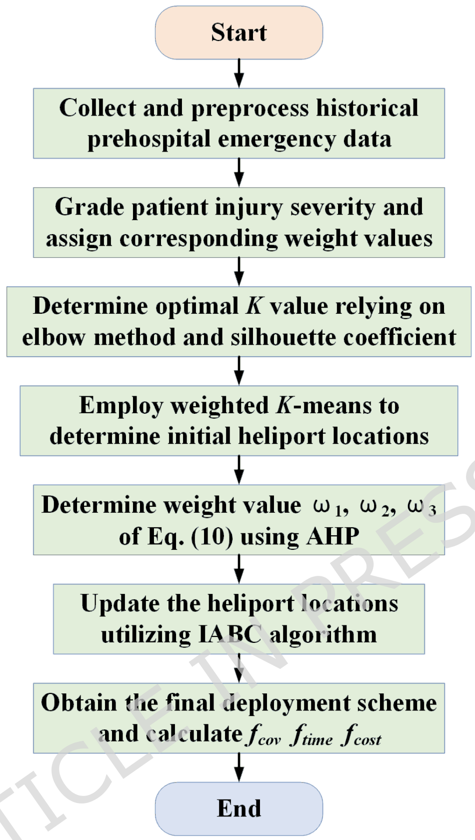

Figure 1. The flowchart of WKM-IABC.

2.4.2 Weight Values Determination

The determination of the weight coefficients $\omega_1, \omega_2, \omega_3$ of the three sub-objectives is the key to the multi-objective integrated optimization model, which directly affects the optimization result and the practical application value of the algorithm. To avoid the subjectivity and arbitrariness of manual weight assignment, this paper uses the Analytic Hierarchy Process (AHP) combined with expert opinions to scientifically determine the weight coefficients of the three sub-objectives[38].

The multi-objective integrated optimization model based on the AHP weighted sum method adopted in this paper is a targeted design closely combined with the actual management scenario of prehospital emergency, which effectively solves the problem of conflicting multi-objectives in heliport deployment and improves the practical feasibility of the optimization result. We fully recognize the advantages of multi-objective optimization algorithms based on Pareto dominance in exploring the target space and providing multiple trade-off solutions. Therefore, we will explore how to extend the integrated optimization model to multi-objective Pareto optimization in future research, in order to provide richer decision support information for emergency management departments.

2.5. Algorithm Flow

The WKM-IABC algorithm flow is shown in Figure 1 . The specific process is described as follows:

Step 1: Collect the historical pre-hospital emergency data and preprocess them;
Step 2: Grade the patient injury severity and assign the corresponding weight values;
Step 3: Determine the optimal number of heliports K relying on elbow method and silhouette coefficient;
Step 4: Employ weighted K-means to determine the initial heliport locations;

Step 5: Use AHP method to determine the $\omega_1, \omega_2, \omega_3$ weight value of Eq. (10) based on expert experience.

Step 6: Utilize the IABC algorithm to update the initial locations of the heliports until the maximum number of iterations;

Step 7: Obtain the final deployment scheme and calculate the service coverage ratio, rescue response time and economic cost.

2.6 Pseudocode of WKM-IABC

Based on the above research, the pseudocode of WKM-IABC is designed as follows:

Input: Emergency demand point set D, patient injury weight ω_j, helicopter service radius R, flight speed v, cost parameters C_{fix} and C_{var}, ABC algorithm parameters SN, $MaxItr$, $Limit$, α, δ.
Output: Optimal heliport deployment Loc, comprehensive fitness F_{best}, service coverage ratio f_{cov}, rescue response time f_{time} and economic cost f_{cost}

```
1. Clean and normalize D;
2. Grade patient injury severity and assign weight ω_j;
3. Determine optimal K value using elbow method and silhouette coefficient;
4. Run Weighted K-means to get initial cluster centers
5. Use AHP to obtain objective weights ω_1, ω_2, ω_3;
6. Initialize ABC honey source population;
7. Calculate F of each honey source using Eq.(10);
8. For iter = 1 to MaxItr
     For i = 1 to SN
       If rand < 0.7
       μ_new = μ_best + α × (μ_best − μ_i)
       Else
       μ_new = μ_i + rand × (μ_i − μ_{i−1})
       Calculate F_new using Eq.(10)
         If F_new > F_i
         μ_i = μ_new
         Else
         stagnation count + 1
         End
       End
     End
     For i = 1 to SN
       If stagnation count > Limit
       μ_i = Cent_k + δ × R × rand
       Reset stagnation count, calculate F_new
       End
     End
     If F_max > F_best
     F_best = F_max
     Set optimal heliport location set Loc
     End
   End
9. Calculate f_cov, f_time and f_cost
10. Output Loc, F_best, f_cov, f_time and f_cost
```

2.7 Computational Complexity Analysis

The WKM-IABC algorithm is a hybrid algorithm combining weighted K-means and IABC, and its total computational complexity is the sum of the complexity of the two parts, and there is no nested iteration between the two parts, so the complexity is linearly additive. Computational complexity of weighted K-means: Assuming that the number of emergency demand points is M, the number of cluster centers is K, and the number of iterations required for the weighted K-means algorithm to converge is t. In each iteration, the algorithm needs to calculate the weighted Euclidean distance between each demand point and each cluster center, with a complexity of $O(KM)$. Therefore, the total computational complexity of the weighted K-means part is $O(t \times KM)$. Computational complexity of IABC: Assuming that the number of honey sources in the

ABC algorithm is SN, the maximum number of iterations is $MaxItr$, and the number of coordinate dimensions of the heliport location is D. In each iteration, a single iteration complexity is $O(SNt \times D)$. Therefore, the total computational complexity of IABC part is $O(MaxItr \times SN \times D)$. The total time complexity is $O(t \times KM + MaxItr \times SN \times D)$.

3. Results

To comprehensively verify the performance of the WKM-IABC algorithm proposed in this paper, we conduct a series of simulation experiments, including optimal K value analysis, comparative experiment with advanced algorithms, ablation experiment, sensitivity analysis, computational complexity analysis and statistical significance test. All the experiments are conducted in the same software and hardware environment to ensure fairness and reliability.

3.1 Experiment Environment

3.1.1 Experimental Data and Problem Parameters

According to the population density distribution in Xi'an, China, 3000 sets of historical patients are randomly generated as the sample points for the experiment. The population density distribution data of Xi'an is derived from publicly available official statistical information in China[39]. The outline information about Xi'an in Figure 2 and Figure 4 comes from the web "China National Platform for Common Geographic Information Services"(https://www.tianditu.gov.cn/about/copyright), and then drawn by MATLAB 2021b. In addition, the 3000 sets of historical patient data used in this case study are entirely synthetic data generated through computer programs, intended solely for method validation and numerical experiments. Then, these sample points are divided into 4 levels including mild patients, moderate patients, severe patients and critical patients relying on the severity of injury, and then they are given a weight value of 1 to 4. The proportions of each level in the total number of patients are 50%, 30%, 15% and 5%, respectively. To visually represent the distribution characteristics of the historical patients, a scatter plot of the 3000 sample points is presented in Figure 2.

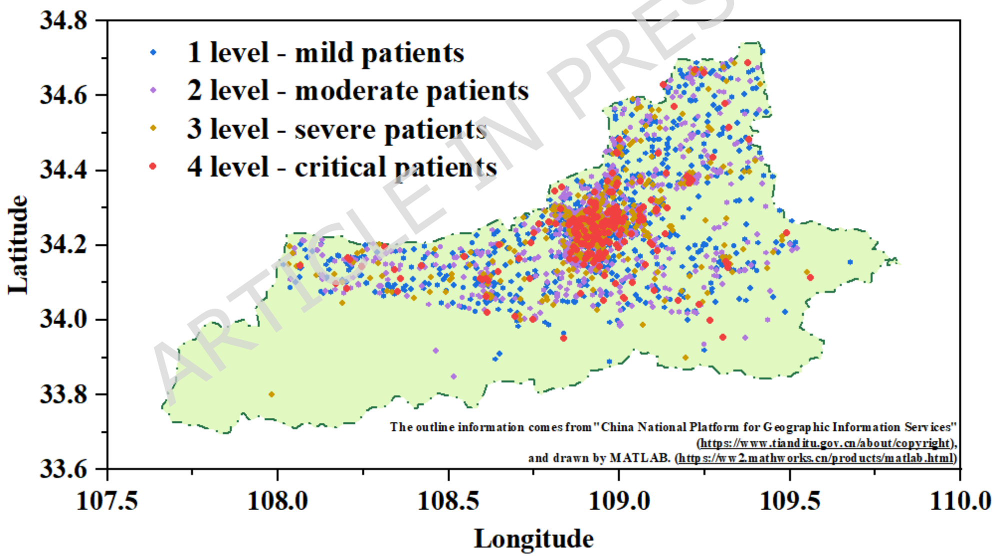

Figure 2. The distribution of the historical patients.

In addition, we assume that the service radius and flight speed of each helicopter are uniform in the experiment, set at 10000 meters and 200 km/h. The construction and operational cost for single heliport C_{fix} is estimated at 1000000 yuan, the variable flight cost is 500 yuan per kilometer. The weight values of the comprehensive single-objective optimization function (Eq. (10)) are determined according to the AHP method described in Section 2.4.2, that is $\omega_1 = 0.5, \omega_2 = 0.4, \omega_3 = 0.1$.

3.1.2 Comparative Algorithms and Parameter Settings

To demonstrate the superiority of the WKM-IABC algorithm, we selected the representative optimization algorithms for comparison, including the original four algorithms, three state-of-the-art algorithms, and conducted ablation experiments. The specific algorithm introduction and parameter design are as follows:

- **WKM-IABC:** the number of employed bees and onlooker bees is 50, the number of honey sources is $SN = 50$, the maximum iteration number $MaxItr = 300$, the abandonment threshold $Limit = 50$, the global optimal learning probability $P_{learn} = 0.7$, the selection pressure coefficient $\alpha = 4.0$, and the disturbance radius $\delta = 0.1R$.

- **Multi-Objective Genetic Algorithm II (NSGA-II)[40]:** the population size is 100, the maximum iteration number $MaxItr = 300$, simulated binary crossover (SBX) probability is 0.9, distribution index $\eta_c = 20$, polynomial mutation probability is $1/D$, D is the variable dimension, and distribution index $\eta_m = 20$.
- **Multi-Objective Particle Swarm Optimization (MOPSO)[41]:** the population size is 100, the maximum iteration number $MaxItr = 300$, the inertia weight linearly decreasing from 0.9 to 0.4, individual learning factor $c_1 = 1.5$, social learning factor $c_2 = 1.5$, the external archive size is 200, and the number of grid divisions used for density estimation set to 30.
- **Grey Wolf Optimization (GWO)[42]:** the population size is 50, the maximum iteration number $MaxItr = 300$, and the control parameter a linearly decreases from 2 to 0.
- **Sparrow Search Algorithm (SSA)[43]:** the population size is 50, the maximum iteration number $MaxItr = 300$, the discoverer ratio is 70%, the alert ratio is 20%, and the safety threshold $ST = 0.8$.
- **Improved Sparrow Search Algorithm (ISSA)[44]:** the population size is 50, the maximum iteration number $MaxItr = 300$, the discoverer ratio is 70%, the alert ratio is 20%, and the safety threshold $ST = 0.8$, the step size is adaptive.
- **Grey Wolf Optimization and Ant Colony Optimization (GWO-ACO)[45]:** the population size is 50, the maximum iteration number $MaxItr = 300$, the control parameter a linearly decreases from 2 to 0, ACO pheromone evaporation factor is 0.1.
- **Adaptive Particle Swarm Optimization (APSO)[46]:** the population size is 50, the maximum iteration number $MaxItr = 300$, tthe inertia weight is adaptive, individual learning factor $c_1 = 1.5$, social learning factor $c_2 = 1.5$.

To ensure fairness in comparison, all the multi-objective integrated optimization models have the same weighted single-objective function (Eq. (10)) as the fitness function. As for NSGA-II and MOPSO, they are designed for Pareto optimization. The maximum iteration number is set to 300 to eliminate the impact of different iteration times. Each algorithm is independently run 30 times on the experimental dataset to eliminate the influence of randomness.

Finally, to ensure the accuracy of the experimental results, the codes involved in this simulation case are written in MATLAB language and run on a computer with Intel Core I7-10710U processor, 1.61 GHz main frequency, 16.0 GB memory and windows 10 (64 bit) operating system.

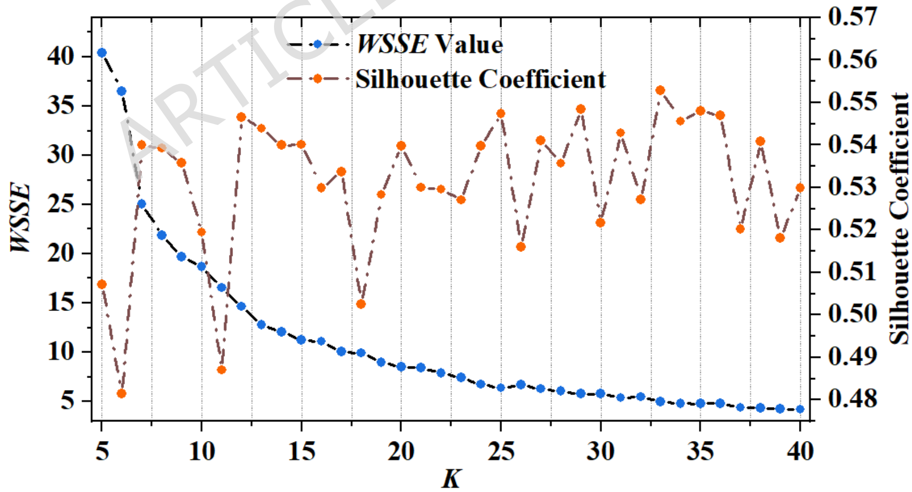

Figure 3. The plot of $WSSE$ value and silhouette coefficient.

3.2 Optimal Number of Heliports Analysis

As shown in Figure 3, we simultaneously plotted the curves of $WSSE$ values and average silhouette coefficients of the weighted K-means algorithm as a function of candidate K values (from 5 to 20). The $WSSE$ curve presents a typical elbow shape: when the K value is less than 12, $WSSE$ decreases sharply with increasing K; when the K value exceeds 12, the downward trend significantly slows down, forming a clear elbow point. At the same time, the average silhouette coefficient reached a local peak of 0.56 at $K = 12$, indicating that the tightness within the cluster and the separation between clusters reached the optimal balance at this time. Considering the elbow method and the principle of maximizing silhouette coefficients, the optimal number of stopping points for this experimental scenario is determined to be $K = 12$. Therefore, all the subsequent algorithm comparisons and optimizations are based on this optimal number of heliports.

3.3 Comparative Analysis with Advanced Algorithms

Table 1 shows the experimental results of various algorithms on five evaluation metrics. The robustness is characterized by the standard deviation of the service coverage ratio of 30 independent runs, and the smaller the value, the better the robustness of the algorithm. Obviously, the WKM-IABC algorithm proposed in this paper has the best performance in service coverage ratio, convergence accuracy, and robustness compared to the other algorithms, with a 11.11% improvement in coverage ratio over the second ranked GWO algorithm, and the traditional multi-objective algorithms (NSGA-II, MOPSO) cannot achieve satisfactory service coverage ratio. This confirms that it is correct for us to consider service coverage as the most important constraint based on expert experience and AHP method in this study. It is worth noting that the rescue response time of WKM-IABC is not the smallest but still acceptable, and the economic cost is slightly higher than other algorithms. The main reason is WKM-IABC takes the maximization of weighted service coverage ratio as the core optimization goal. In order to realize the full coverage of the demand points in the edge area and the critical patients in the low population density area, some heliports need to be deployed in the edge area that will be ignored by the traditional algorithm. The increase of economic cost is mainly because the expansion of coverage leads to a slight increase in the average flight distance of helicopters, which brings a small increase in variable operating cost. However, compared with other algorithms, WKM-IABC realizes the core goal of exchanging a small increase in response time for full coverage of critical patients in prehospital emergency, and from the perspective of emergency resource utilization efficiency, the cost-benefit ratio of WKM-IABC is still acceptable.

	Service Coverage Ratio	Rescue Response Time (s)	Economic Cost (W)	Convergence Accuracy	Robustness (SD)
NSGA-II	72.48%	36.64	1604.78	0.412	0.028
MOPSO	71.20%	44.07	1678.23	0.405	0.031
GWO	75.18%	52.48	1801.23	0.438	0.019
SSA	71.71%	58.07	1800.92	0.410	0.022
ISSA	79.77%	55.21	1813.61	0.445	0.017
GWO-ACO	80.83%	56.38	1858.02	0.452	0.015
APSO	82.74%	57.12	1877.39	0.461	0.013
WKM-IABC	86.29%	58.66	1971.48	0.482	0.007

Table 1. The comparative experimental results on various evaluation metrics.

To compare the service coverage ratio of helicopter prehospital emergency service more intuitively to 3000 sets of patients under different algorithms, Figure 4 shows the distribution of heliports location. It can be seen from Figure 4 that each heliport location has a massive change under different optimization algorithms but within the map range. The WKM-IABC algorithm can serve 3000 sets of patients to the maximum extent, especially for critically ill patients. This further indicates that the WKM-IABC algorithm has the highest service coverage and can serve more patients.

3.4 Ablation Experiment Results Analysis

To analyze the specific contributions of each component in the WKM-IABC algorithm, we designed ablation experiments and compared the performance of five algorithms, including K-means, weighted K-means, traditional ABC, IABC, and WKM-IABC algorithms.

	K-means	Weighted K-means	Traditional ABC	Improved ABC	WKM-IABC
Service Coverage Ratio	81.01%	81.54%	85.20%	86.15%	86.29%
Rescue Response Time (s)	47.35	46.6	57.78	59.27	58.66
Economic Cost (W)	1784.58	1779.14	1950.34	1978.24	1971.48
Convergence Speed	–	–	268	215	189

Table 2. The ablation comparative experimental results on various evaluation metrics.

Table 2 shows the experimental results of five algorithms on various evaluation metrics. Firstly, by comparing the experimental results of two clustering algorithms, it can be seen that by grading the severity of patient injury of the historical

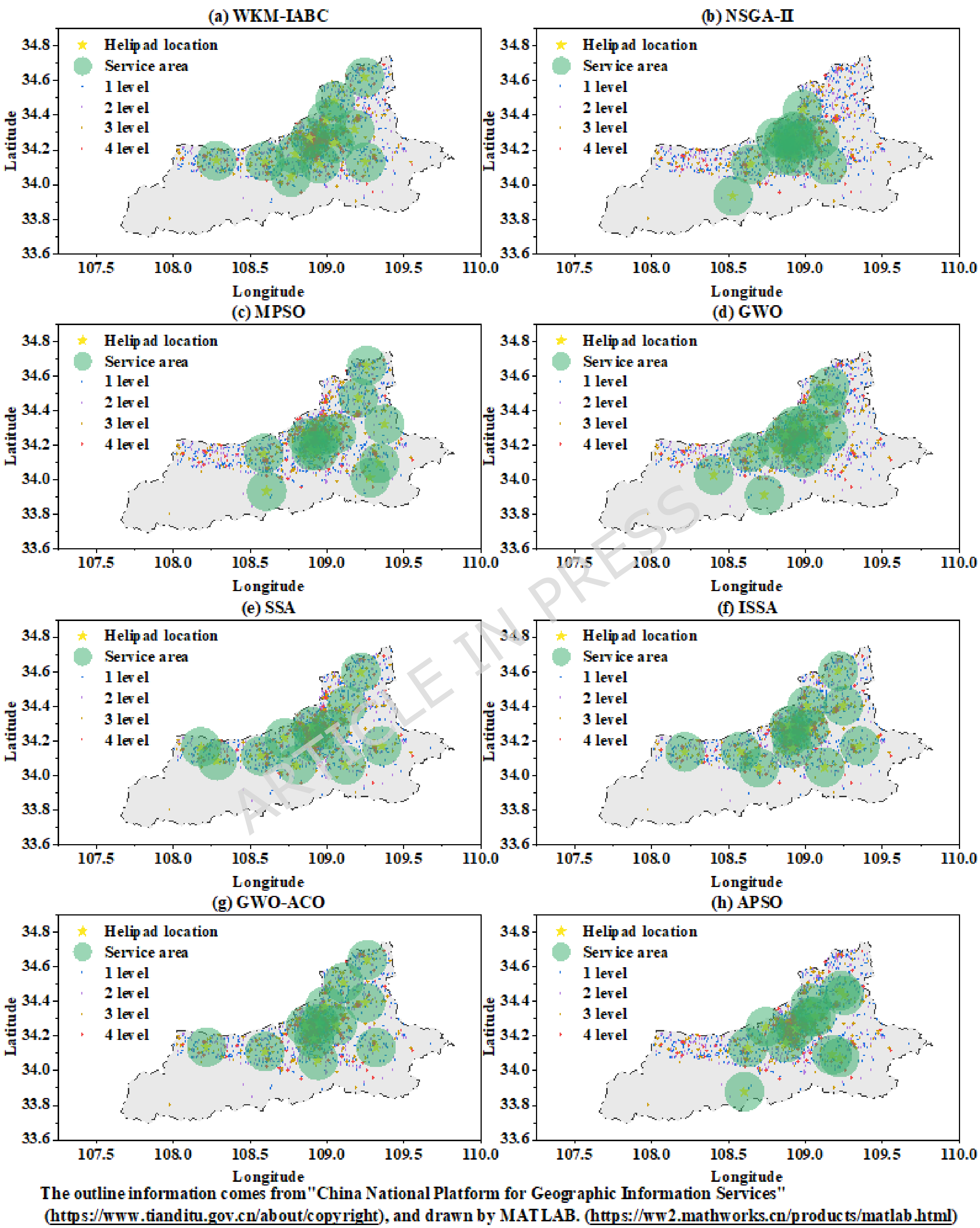

Figure 4. The distribution of heliports locations under different algorithms.

data and giving different weights, the heliports locations can be designed more reasonably, so as to improve the service coverage ratio. This indicates that introducing weight vectors based on the severity of patient injury is beneficial for the reasonable deployment. Secondly, the deployment results of the standard ABC algorithm and IABC algorithm are basically the same, and IABC has a higher coverage ratio, which indicates that the ABC algorithm is an optimization algorithm suitable for solving the deployment of prehospital emergency heliports problem, and the improvement strategy can effectively improve the algorithm performance. Finally, it is easy to see that the algorithm proposed in this article achieved the best overall performance, which further proves the superiority of the hybrid intelligent optimization algorithm in balancing multiple constraint conditions.

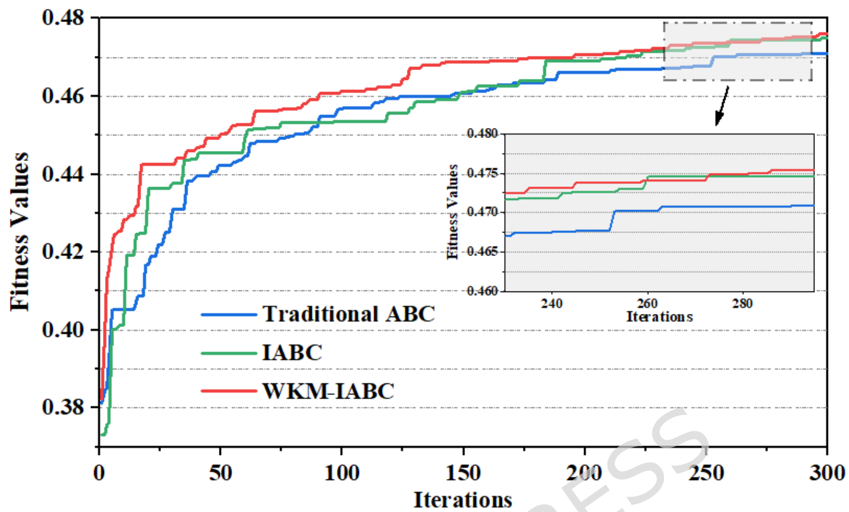

Figure 5. The fitness convergence curve of the traditional ABC, IABC, and WKM-IABC algorithm.

To demonstrate the faster convergence speed of the WKM-IABC algorithm proposed, Figure 5 shows the fitness convergence curves of the traditional ABC, IABC, and WKM-IABC algorithm. It can be seen that the fitness convergence speed of the WKM-IABC algorithm is significantly faster than that of the other two. This is because the weighted K-means provides a more accurate initial solution and determines the correct search direction more quickly, while the initial solution of the traditional ABC algorithm is random and of poor quality. In addition, it is not difficult to find that the convergence result of the final fitness of the hybrid intelligent optimization algorithm is optimal, which proves that the WKM-IABC algorithm can achieve the best heliports deployment scheme.

3.5 Sensitivity Analysis

Sensitivity analysis was conducted for three core adjustable parameters of the WKM-IABC algorithm to verify the stability. The single factor variable method was adopted: only one parameter was changed while other parameters were fixed, and the change trends of the three core evaluation indexes were analyzed.

3.5.1 Sensitivity Analysis of AHP Weight Coefficients

The AHP weight coefficients determine the optimization priority of the algorithm, and the core variable is the service coverage ratio weight was adjusted from 0.3 to 0.7 with a step of 0.1, and the experimental results are shown in Figure 6 (a). With the increase of ω_1, the service coverage ratio shows a significant increasing trend, rescue response time shows a slow increasing trend, and the economic cost shows a gradual increasing trend, which is consistent with the optimization priority setting, indicating that the algorithm can effectively respond to the decision preferences of emergency management departments. When $\omega_1 = 0.5$, the three indexes reach the best practical balance, which verifies the rationality of the original weight setting.

3.5.2 Sensitivity Analysis of Helicopter Service Radius

The service radius is the key technical parameter, which directly affects the coverage capacity and rescue efficiency of helicopters. It was adjusted from 6000 to 14000 with a step of 2000, and the experimental results are shown in Figure 6 (b). With the increase of service radius, the service coverage ratio, the rescue response time, and the economic cost show a rapid increasing trend. When service radius is 10000, the growth rate slows down, which verifies that the original setting is the optimal technical parameter for the experimental scenario.

3.5.3 Sensitivity Analysis of Optimal Cluster Number

The cluster number K is the optimal number of heliports, which is the core decision parameter of the deployment scheme. K was adjusted from 8 to 16 with a step of 2, and the experimental results are shown in Figure 6 (c). With the increase of K , the service coverage ratio shows a gradual increasing trend, the rescue response time shows a significant decreasing trend, and the economic cost shows a significant increasing trend. When $K > 12$, the growth rate slows down, which is consistent with the elbow method result in Section 3.2, further verifying that $K = 12$ is the optimal number.

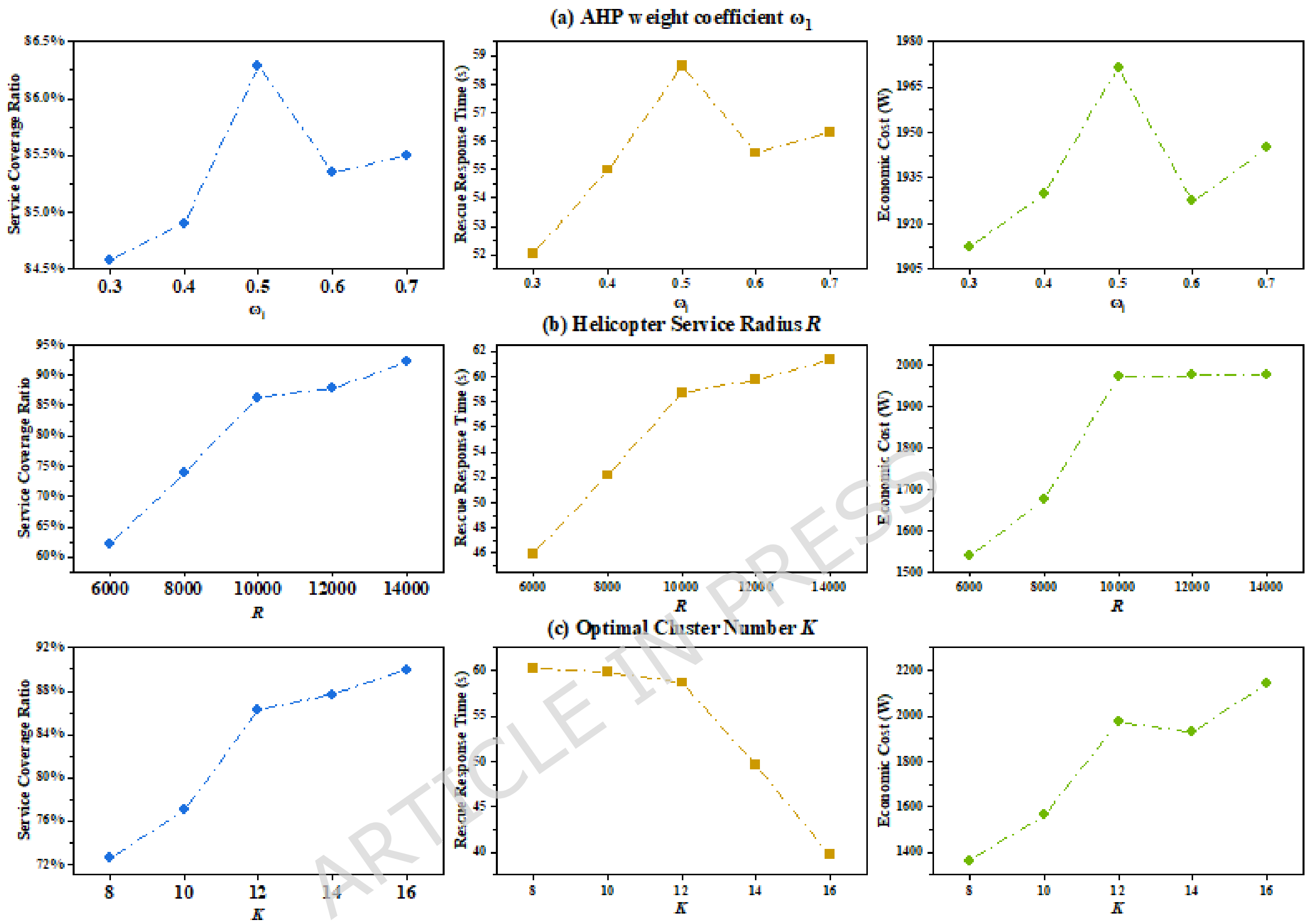

Figure 6. The distribution of heliports locations under different algorithms.

3.6 Complexity Comparison with Other Algorithms

To further verify the computational efficiency of the WKM-IABC algorithm, we conduct a comprehensive comparison with all comparative algorithms. The comparison results are shown in Table 3. Although WKM-IABC shares the same complexity order as Traditional ABC, GWO, and SSA, its weighted K-means initialization reduces the effective $MaxItr$. This makes its actual runtime shorter than the comparative algorithms. WKM-IABC achieves a balance between complexity and performance: it retains the low linear complexity of classic intelligent algorithms, and leverages high-quality initialization to outperform all comparative algorithms in practical runtime. This confirms its suitability for efficient optimization of prehospital emergency heliport deployment.

3.7 Statistical Significance Tests

To verify the statistical reliability of WKM-IABC algorithm, eliminating the impact of randomness in intelligent optimization, paired sample t-tests were conducted on the 30 independent run results of WKM-IABC and all comparative algorithms. The test focused on the three core evaluation indexes (Service Coverage Ratio, Convergence Accuracy, Robustness). $P<0.05$ indicates a significant difference; $P<0.01$ indicates an extremely significant difference. The test results are shown in Table 4.

For all three core indexes, WKM-IABC has extremely significant statistical differences compared with every comparative algorithm. This indicates that the algorithm's advantages in service coverage ratio, convergence accuracy, and stability are not

Algorithm	Computational Complexity	Key Notes
NSGA-II	$O(MaxItr \times N^2 \times D)$	Quadratic complexity due to non-dominated sorting
MOPSO	$O(MaxItr \times N \times D + N \times A)$	A = archive size (200), higher than basic PSO
GWO	$O(MaxItr \times N \times D)$	Linear complexity, no additional overhead
SSA	$O(MaxItr \times N \times D)$	Similar to GWO, simple population update
ISSA	$O(MaxItr \times N \times D)$	ISSA with adaptive step, same complexity order
GWO-ACO	$O(MaxItr \times (N \times D + N^2))$	ACO pheromone matrix adds $O(N^2)$ overhead
APSO	$O(MaxItr \times N \times D)$	Adaptive inertia weight, no complexity increase
ABC	$O(MaxItr \times SN \times D)$	Same order as ABC variants, random initialization
IABC	$O(MaxItr \times SN \times D)$	Three improvement strategies, same complexity order
WKM-IABC	$O(tKM + MaxItr \times SN \times D)$	Hybrid design, low effective complexity

Table 3. The computational complexity comparison.

Index	Service Coverage Ratio		Convergence Accuracy		Robustness	
Algorithm	t-value	P-value	t-value	P-value	t-value	P-value
NSGA-II	28.63	<0.001	32.15	<0.001	-22.58	<0.001
MOPSO	31.25	<0.001	34.72	<0.001	-24.31	<0.001
GWO	21.56	<0.001	24.89	<0.001	-18.32	<0.001
SSA	29.87	<0.001	30.51	<0.001	-19.65	<0.001
ISSA	18.42	<0.001	22.67	<0.001	-13.78	<0.001
GWO-ACO	16.35	<0.001	20.18	<0.001	-12.46	<0.001
APSO	15.32	<0.001	18.65	<0.001	-12.65	<0.001
ABC	9.68	<0.001	12.34	<0.001	-8.92	<0.001
IABC	4.18	0.002	5.25	<0.001	-5.18	<0.001

Table 4. Paired sample t-test results (WKM-IABC vs. all comparative algorithms).

due to random fluctuations in experimental results, but are inherent advantages brought by its hybrid optimization design. In a word, the statistical tests fully confirm that the comprehensive superiority of WKM-IABC over state-of-the-art algorithms is statistically rigorous and reliable, providing a solid basis for the algorithm's practical promotion and application.

4. Conclusions

The scientific deployment of prehospital emergency heliports is a core problem for improving the efficiency of aviation medical rescue and optimizing resource allocation. To address this problem, this paper proposes a hybrid intelligent optimization algorithm WKM-IABC. The algorithm innovatively integrates domain knowledge and data-driven initialization, multi-objective optimization based on decision preferences, and multi-dimensional improvement of the ABC algorithm, forming a complete and scientific heliport deployment optimization framework. Simulation experiments, ablation experiments, sensitivity analysis and statistical significance tests show that the WKM-IABC algorithm has significant comprehensive superiority over the state-of-the-art optimization algorithms. The algorithm can realize the optimal balance between the three core objectives of service coverage ratio, rescue response time, and economic cost. However, this study is conducted under static demand and determined parameters, so the future work can be devoted to considering dynamically changing emergency demands, exploring interactive multi-objective optimization methods, and providing a set of Pareto optimal solutions to enhance deployment flexibility[47,48].

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Author contributions statement

Conceptualization, X.H., and Y.L.; methodology, X.H., and D.L.; software, T.Z.; validation, L.X., and Y.L.; formal analysis, X.H., and D.L.; investigation, D.L., and X.H.; resources, Y.L.; data curation, X.H., and D.L.; writing—original draft preparation, X.H., and D.L.; writing—review and editing, X.H., D.L., T.Z., and P.G.; visualization, Y.L., and P.G.; supervision, P.G.; project administration, P.G.; funding acquisition, P.G. All authors have read and agreed to the published version of the manuscript.

Competing interests

The authors declare no competing interests.

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